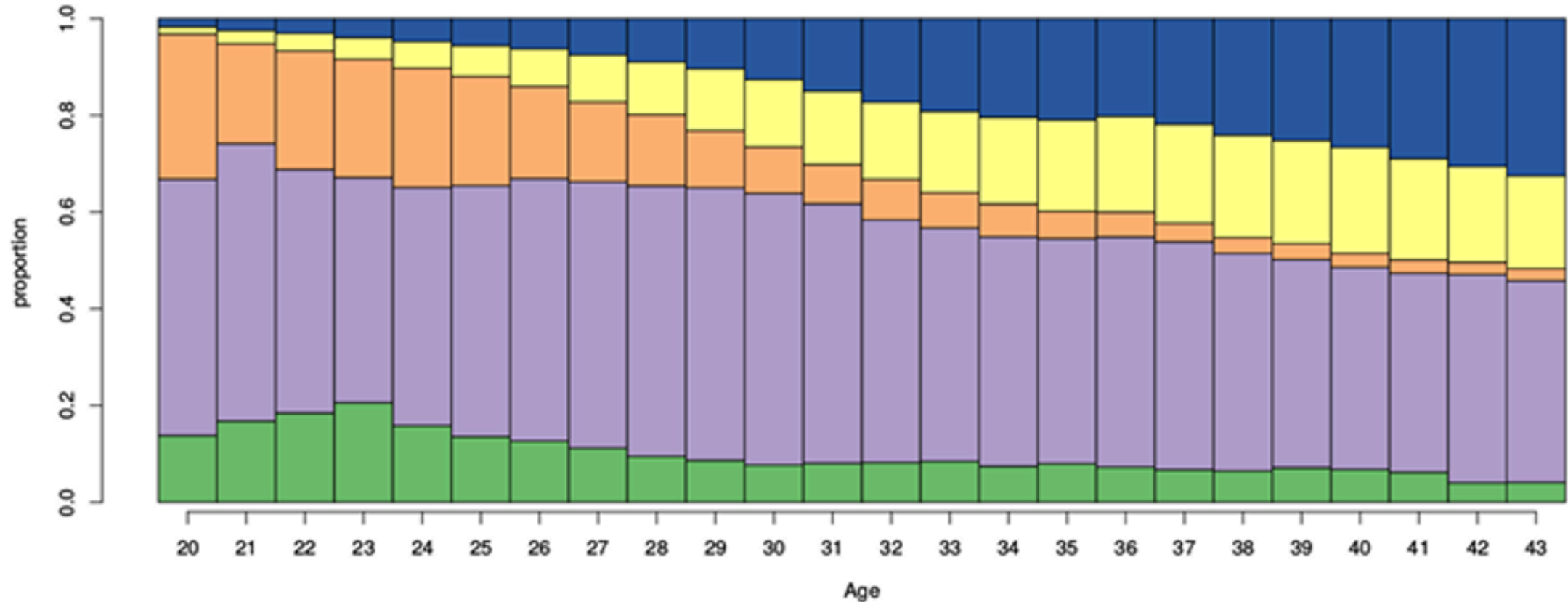


THE ROLE OF CONTEXT IN SHAPING ENTREPRENEURSHIP

LECTURE 3

JEROEN MAHIEU

WHEN DO PEOPLE BECOME SELF-EMPLOYED?



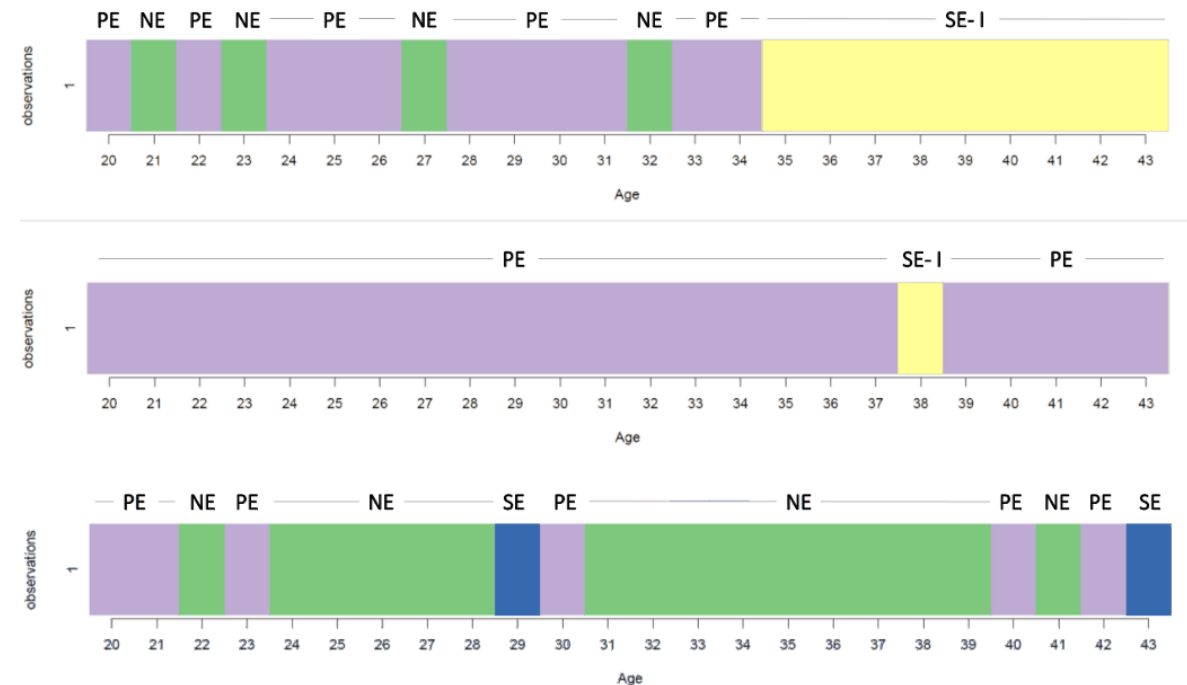
■ Non-Employed (NE) ■ School (SCH) ■ Unincorp. Self-Employment (SEU)
■ Paid Employment (PE) ■ Incorp. Self-Employment (SEI)

Source: Humphries, J. E. (2017). *The causes and consequences of self-employment over the life cycle*. Doctoral dissertation, The University of Chicago. Figure shows the career states of Swedish males born in 1970 who are ever self-employed between 1990 and 2013.

MOST FUTURE ENTREPRENEURS ARE EMPLOYEES FIRST

- At ages 20–25, most people in the sample are still in **school or paid employment**
- Paid employment remains the largest career state throughout the observed age range
- Incorporated and unincorporated self-employment become more common as the cohort gets older
- But entrepreneurship is also transitory: only **45–50%** of those who ever became self-employed are still so by age 43
- Entrepreneurship therefore often looks less like a first job and more like a **career transition**

Figure 2: Three Example Career Trajectories Involving Self-Employment.

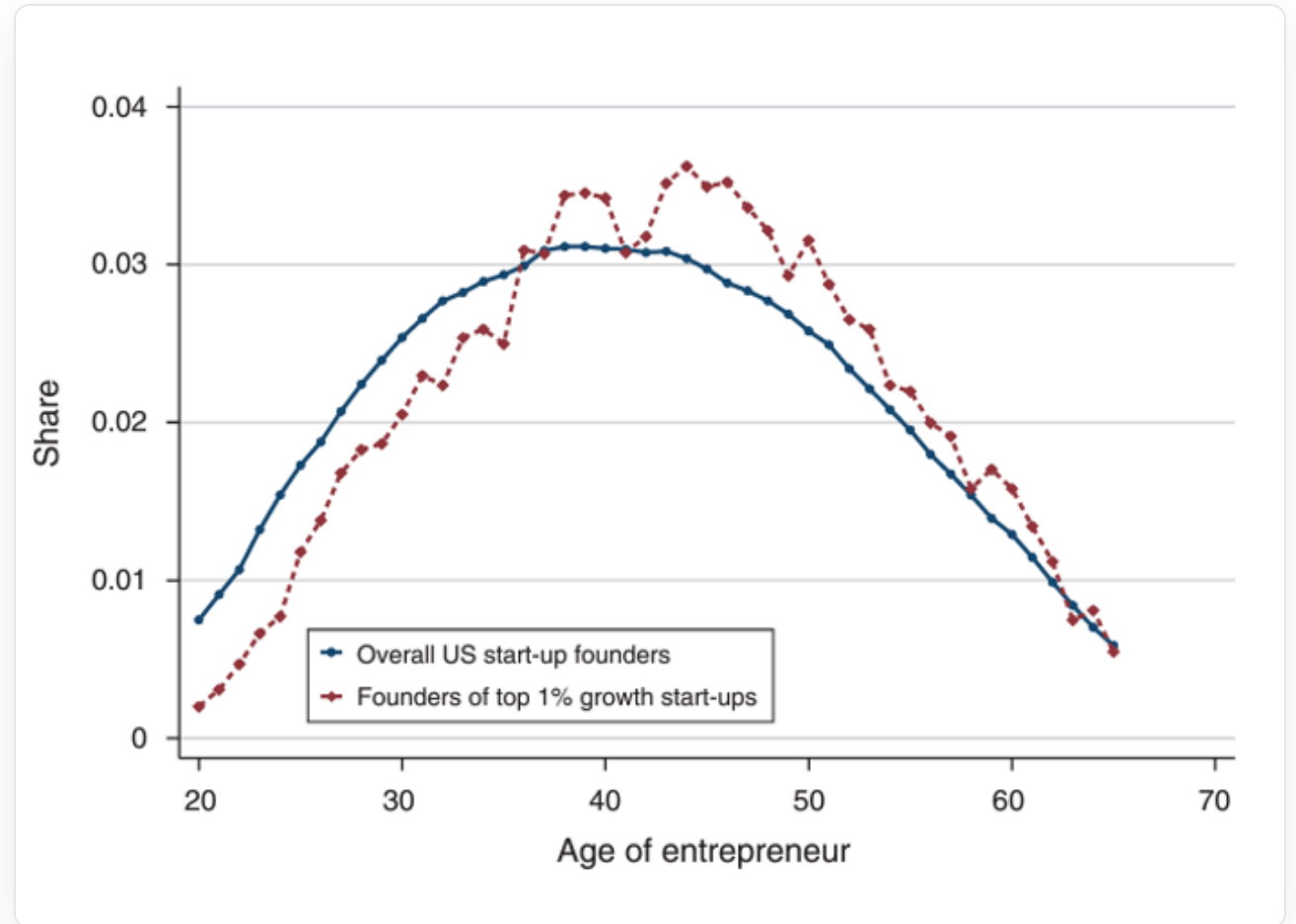


SUCCESSFUL FOUNDERS ARE USUALLY MIDDLE-AGED TOO

Recall the figure from the previous lecture:

- The share of all U.S. startup founders peaks around **age 40**
- The founders of the fastest-growing ventures are older still
- For the **1-in-1,000 fastest-growing ventures**, mean founder age is **45**

The popular image of the 20-year-old founder is memorable but not representative.



ENTREPRENEURS USUALLY COME FROM ORGANIZATIONS

- New ventures are rarely founded by people arriving directly from school or from outside the labor market
- Reviews of the evidence suggest that **at least nine in ten entrepreneurs have previously worked for another organization**
- Their current employer is therefore part of the entrepreneurial process, even before a new firm exists

WHAT THE ORGANIZATION PROVIDES	HOW IT MAY MATTER
Skills and knowledge	Technical, commercial, managerial and regulatory knowledge; exposure to unmet customer needs, technology and rejected projects
Beliefs and values	Demystify entrepreneurship, reduce stigma, spur a taste for autonomy
Relationships	Future cofounders, employees, customers, suppliers and investors
A career alternative	Promotion prospects and wages determine the cost of leaving

The same employer may simultaneously *prepare people to leave* and *give them reasons to stay*.

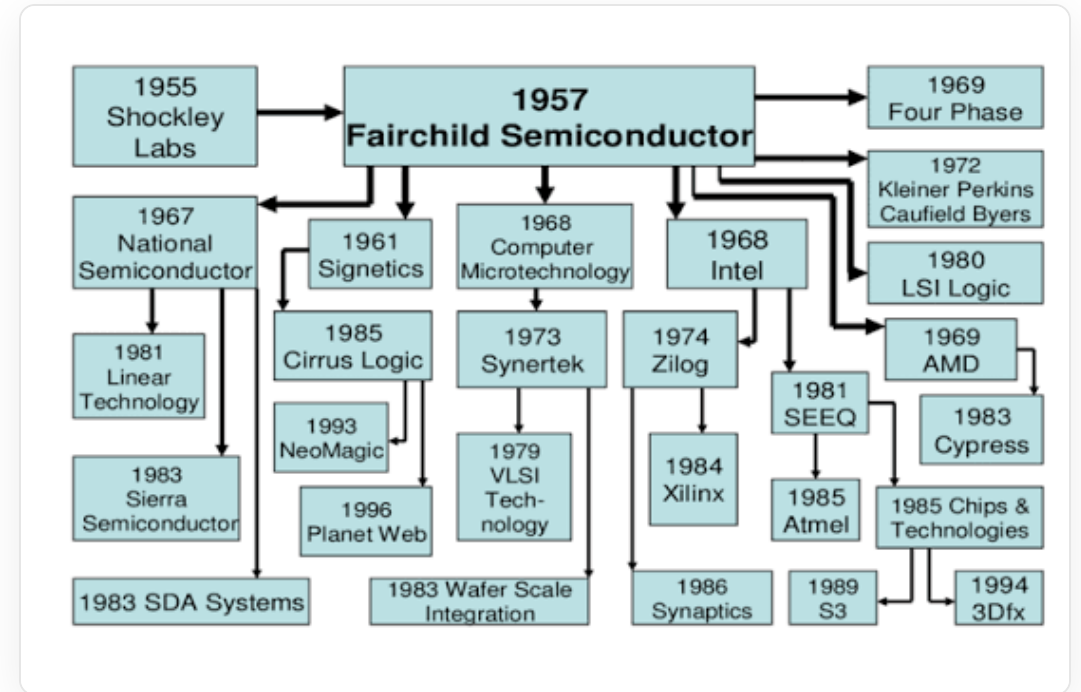


THE TRAITOROUS EIGHT

Eight employees who left **Shockley Semiconductor Laboratory** in 1957—only two years after its founding—to found **Fairchild Semiconductor**. This eventually led to numerous spinoffs, including AMD and Intel. Over 20 years, **65 different companies** were started by 1st- or 2nd-generation teams that traced their Silicon Valley origins to Shockley Semiconductor. In 2014, an estimated 92 of 130 listed descendant firms were together worth over **US\$2.1 trillion**.

Why did so much entrepreneurship emerge from one organizational community?

- Did Shockley **select** unusually entrepreneurial people?
- Did working there **develop** valuable capabilities?
- Did alumni keep helping one another through a powerful **network**?



A MAP FOR THE EVIDENCE

	PERFORMANCE-NEUTRAL	PERFORMANCE-ENHANCING
SELECTION	SELECTION × NEUTRAL The same preferences that made people join Shockley Labs made them eventually enter entrepreneurship	SELECTION × ENHANCING Entrepreneurial ability influences the choice to join Shockley, and eventually entrepreneurship
TREATMENT	TREATMENT × NEUTRAL Context encourages entry, without demonstrated capability gains	TREATMENT × ENHANCING Context develops resources or capabilities that improve outcomes

SHOCKLEY RECRUITED A STELLAR TEAM

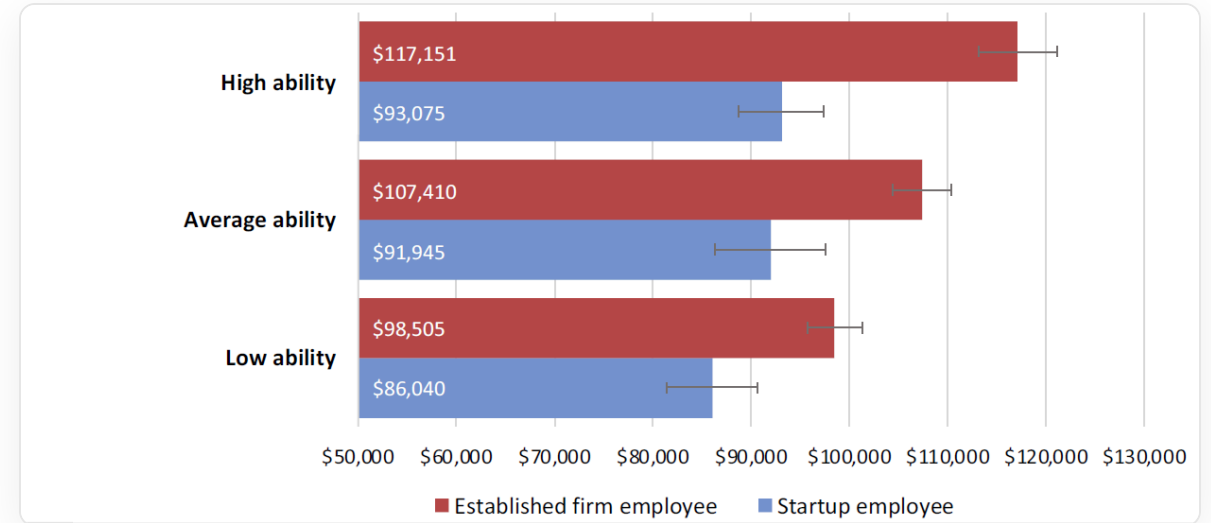
NAME AND BIRTH YEAR	EXPERTISE SHOCKLEY RECRUITED	SELECTED LATER CONTRIBUTION AT FAIRCHILD
Julius Blank, 1925	Mechanical engineering and machining. BA, City College of New York (1950)	Built facilities and production equipment
Victor Grinich, 1924	Electrical engineering, electronics. PhD, Stanford University (1953)	Established device testing, characterization and applications
Jean Hoerni, 1924	Physics and semiconductor theory. PhD, University of Geneva (1950) and Cambridge University (1952)	Invented the planar process
Eugene Kleiner, 1923	Mechanical and production engineering. MA, New York University (1950)	Developed production capabilities; connected the group to financier Arthur Rock
Jay Last, 1929	Optics and solid-state physics. PhD, MIT (1956)	Advanced photolithography; led the first planar integrated-circuit team
Gordon Moore, 1929	Physical chemistry and materials. PhD, California Institute of Technology	Led process research and later Fairchild's R&D
Robert Noyce, 1927	Physics and semiconductor devices. PhD, MIT (1953)	Became general manager; conceived the monolithic integrated circuit
Sheldon Roberts, 1926	Metallurgy and materials science. PhD, MIT (1952)	Established silicon-crystal production

THE YOUNG AND TALENTED?

- Keep in mind that Shockley Semiconductor Laboratory was still a **small startup**: two years old, about 30 employees
- How could Shockley attract these highly able young scientists (**6 of the 8 had a PhD**) with only the few resources he had? Can this in some way also explain their eventual exit to found their own company?

“Talented people don’t need to work for you; they have plenty of options. You should ask yourself a more pointed question: Why would someone join your company as its 20th engineer when she could work at Google for more money and more prestige?”

Peter Thiel (2014), *Zero to One*, p. 20



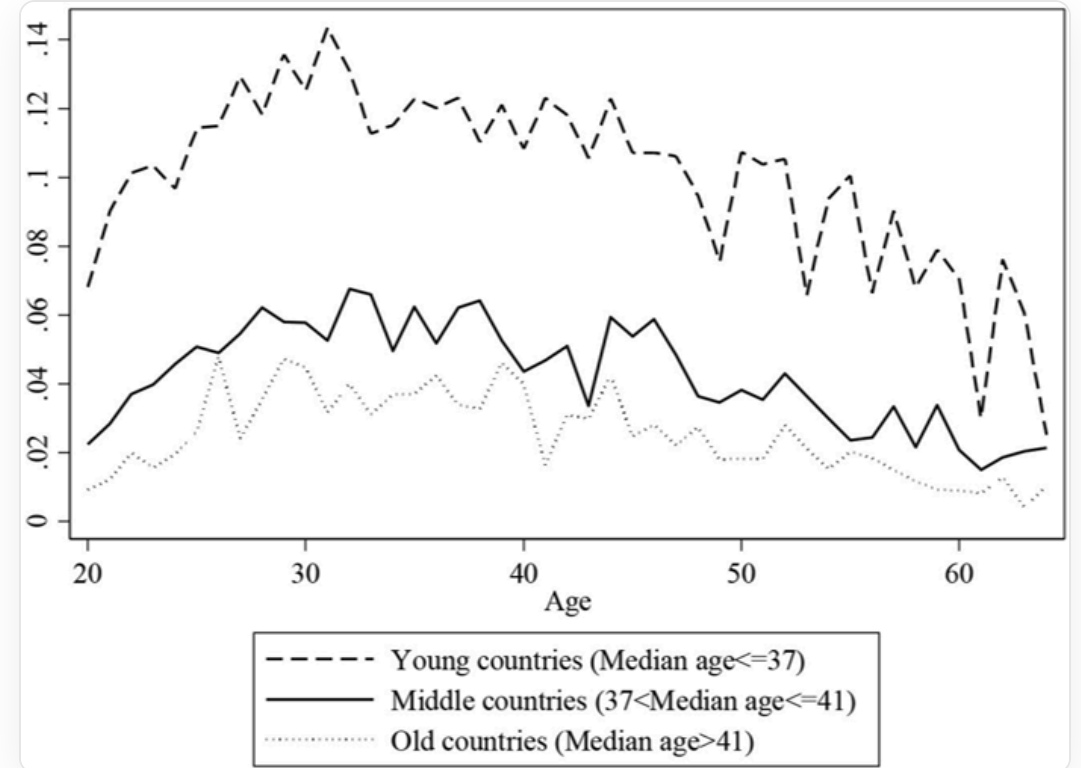
Roach, M., & Sauermann, H. (2024). Can technology startups hire talented early employees? Ability, preferences, and employee first job choice. *Management Science*, 70(6), 3619–3644.

ABILITY AND PREFERENCES

	Startup type: Greater preference benefits from startup employment	Established firm type: Greater preference benefits from established firm employment
High ability worker: Greater productivity benefits from access to resources and complementary assets. In greater demand for cutting-edge R&D projects in both startups and established firms	High ability startup type Stronger trade-off between opportunity costs (forgoing greater productivity benefits from working in established firm) and preference benefits from working in startup Given startup demand for cutting-edge projects: ▪ <i>If preference benefits are greater than opportunity costs, then most likely to work in a startup</i> ▪ <i>If preference benefits are less than opportunity costs, then less likely to work in a startup than low ability startup types</i>	High ability established firm type Stronger reinforcement between productivity benefits and preference benefits from working in established firm Given established firm demand for cutting-edge projects: ▪ <i>Least likely to work in a startup</i>
Low ability worker: Lower productivity benefits from access to resources and complementary assets. In greater demand for incremental R&D projects in established firms	Low ability startup type Weaker trade-off between opportunity costs and preference benefits from working in startup If startups can hire high ability workers: ▪ <i>Less likely to work in a startup than high ability startup types, but more likely than established firm types</i> If startups cannot hire high ability workers: ▪ <i>Most likely to work in a startup</i>	Low ability established firm type Weaker reinforcement between productivity benefits and preference benefits from working in established firm Given established firm demand for incremental projects: ▪ <i>Less likely to work in a startup than startup types</i>

ALSO TO BYPASS THE CAREER LADDER?

- Working in a startup like Shockley Semiconductor Lab also gave these young scientists exposure to **tasks normally reserved for more experienced workers**
- (Inadvertently) prepared them for entrepreneurship? Entrepreneurs need to be creative (inventive) but also need business skills
- Creativity peaks when you are **young**
- Business acumen increases with **experience in high-level positions**
- This can also explain the “inverted-U” relation between entrepreneurship and age (see previous class)
- Implication: having too many older workers in society **slows entrepreneurship**—older workers occupy key positions and block skill acquisition by young people



Liang, J., Wang, H., & Lazear, E. P. (2018). Demographics and entrepreneurship. *Journal of Political Economy*, 126(S1), S140–S196.

TILTING THE ABILITY-PREFERENCES TRADE-OFF

- While Shockley was a brilliant scientist—he won the **Nobel Prize in Physics in 1956**—he was a terrible manager. Instead of inspiring and motivating his hires, he would gradually undermine them and sap their confidence.

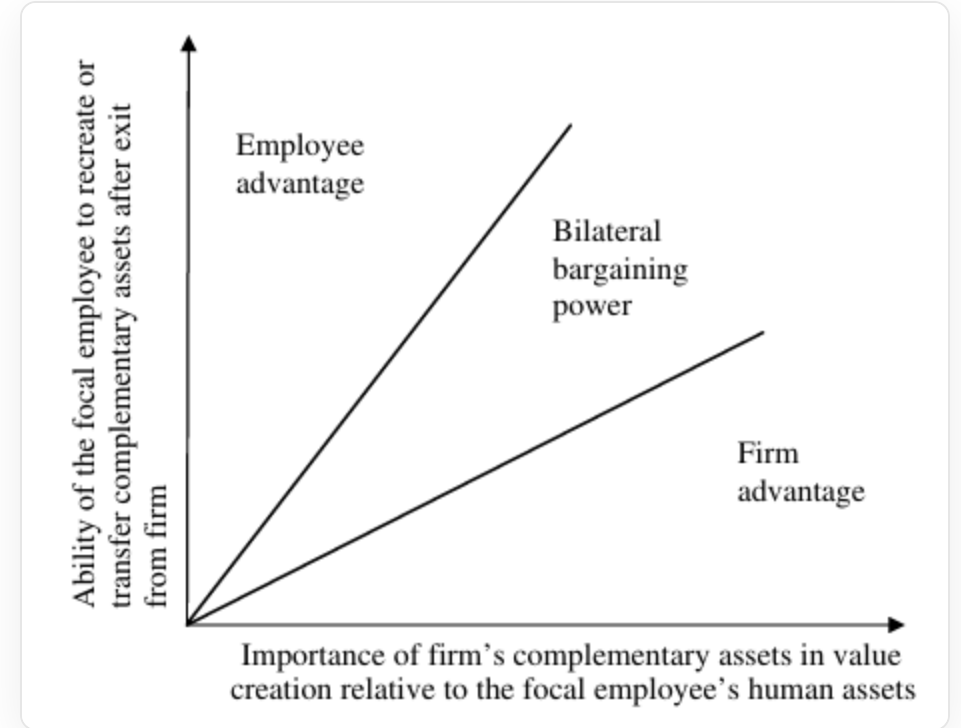
“When he hired you, you were the greatest person in the world. Then slowly you worked your way down the line. First you were brilliant. Then, ‘You’re doing a good job.’ Then, ‘You’re capable, but I’m unsure about you... Now I’m really unsure... Now I think you’re inadequate. I don’t think you can do the job for me.’ He kept a black book on everybody.”

Sheldon Roberts, *Los Angeles Times* (2001)

- Where before *preferences* > *opportunity cost of working elsewhere*, this now shifted—because of the lack of actual challenge, autonomy and creativity
- This will particularly affect the **most able employees**, because for them the trade-off is the steepest

WHO LEAVES? WHERE TO? WHY WORRY?

- The relative bargaining power of firm and employee depends on whether the firm's **complementary assets** (IP, physical capital, reputation, routines, tacit knowledge) are important for value creation *and* whether the employee can walk away with them—or recreate them cheaply after exit
- High-ability workers benefit most from these assets, but are also the ones who can most easily recreate them. Shockley thought he was in the “**firm advantage**” bracket whereas there was actually an “**employee advantage**”
- When these high performers leave, they are more likely to **found a new company** instead of moving to another firm—and to compete with the parent firm



Campbell, B. A., Ganco, M., Franco, A. M., & Agarwal, R. (2012). Who leaves, where to, and why worry? Employee mobility, entrepreneurship and effects on source firm performance. *Strategic Management Journal*, 33(1), 65–87.



Gergely Orosz  @GergelyO... · 40m



All 7 cofounders of Anthropic were OpenAI employees who disagreed with the direction OpenAI was going in

So they started a new company called Anthropic

That was 2021. 5 years later they are the leaders in enterprise AI usage + coding use cases. Safe to say worked out well



MaximeB  @MaximeMB_ · 4h

Just realized Anthropic's CEO was an ex-OpenAI employee 🤖

Experience



CEO and Co-Founder

Anthropic · Full-time
Feb 2021 - Present · 5 yrs 1 mo

Combining scaling and long-term AI alignment methods to build steerable, interpretable, and robust AI systems.



OpenAI

4 yrs 6 mos
San Francisco Bay Area



Vice President of Research

Dec 2019 - Dec 2020 · 1 yr 1 mo

* I led the efforts to build GPT-2 and GPT-3 (<https://arxiv.org/abs/2005.14165>). ...see more



Research Director

Sep 2018 - Dec 2019 · 1 yr 4 mos

WHY PEOPLE LEAVE ALSO MATTERS

- Lee et al. (2024) find that founders who leave their parent firm due to **organizational misalignment motives** (frustration with bureaucracy, strategic disagreement, ...) tend to found successful initial ventures, whereas founders who leave for other reasons rarely do
- This is because founders with organizational misalignment motives engage in “**venture crafting**”—carefully considering and implementing organizational activities to build well-functioning organizations, so that the issues that colored their experiences at their parent firms do not derail their ventures
- Put differently: founders acted deliberately to ensure their ventures were organized *differently* from the firms they had chosen to leave, and chose not to carry certain practices forward

“[...] the idea that you needed something in addition to just scaling the models up, which is alignment or safety. You don’t tell the models what their values are just by pouring more compute into them. And so there were a set of people who believed in those two ideas. We really trusted each other and wanted to work together. And so we went off and started our own company with that idea in mind.”

Dario Amodei, on leaving OpenAI to found Anthropic

DOES BUREAUCRACY BREED ENTREPRENEURSHIP?

If changes in job design and amenities (more routine work, less challenging tasks, top-down decision making) and organizational misalignment drive people to consider entrepreneurship and craft better ventures, shouldn't we expect many entrepreneurs to come from **large, bureaucratic organizations**?

THE FAIRCHILD VIEW

YOUNG ENTREPRENEURIAL FIRMS

- Learn about the entrepreneurial process
- Exposure to a network of suppliers of labor, goods, capital & customers
- Self-selection on risk

THE XEROX VIEW

LARGE BUREAUCRATIC FIRMS

- Incapable of responding to radical innovations that upset established ways of organizing (**inertia**)
- Incapable of evaluating employees' ideas as they fall outside the core business (**myopia**)
- Focus on the core business (the diversification discount)

Which “view” is more prevalent?

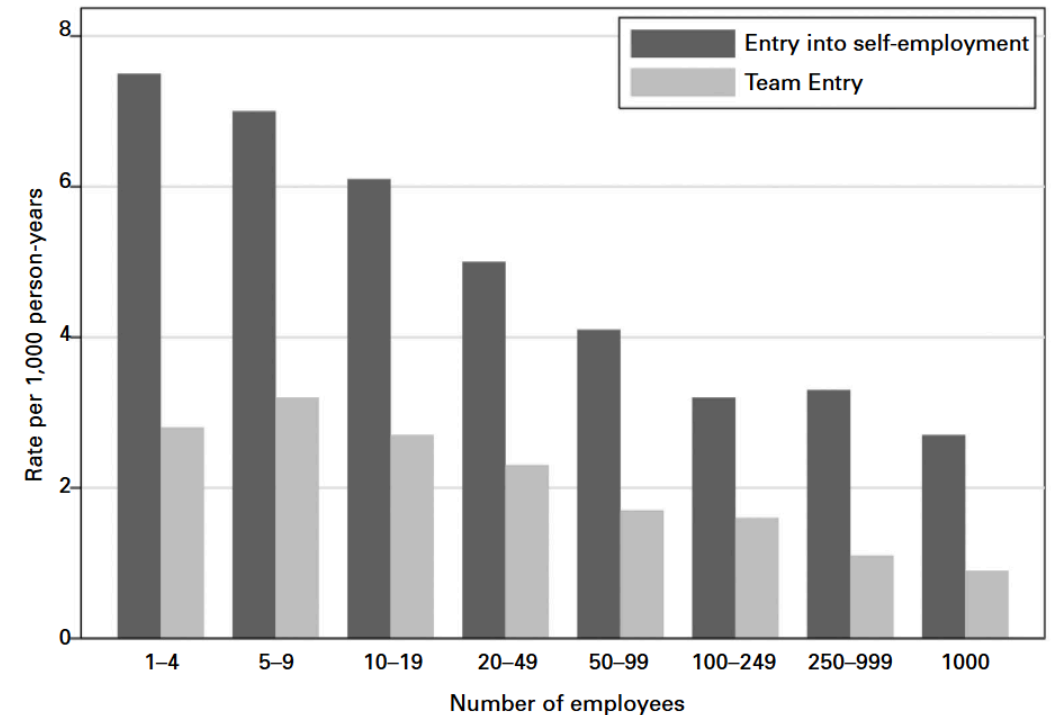
Gompers, P., Lerner, J., & Scharfstein, D. (2005). Entrepreneurial spawning: Public corporations and the genesis of new ventures, 1986 to 1999. The Journal of Finance, 60(2), 577–614.

BUREAUCRATIC EMPLOYERS SPAWN FEWER ENTREPRENEURS

Using Danish employer–employee data, Sørensen (2007) finds:

- Employees of **larger and older organizations** are less likely to enter entrepreneurship
- True for (individual) self-employment *and* for larger startups (multiple joiners in the first year)

Figure 2. Entry into entrepreneurship by size of employer.



Sørensen, J. B. (2007). Bureaucracy and entrepreneurship: Workplace effects on entrepreneurial entry. *Administrative Science Quarterly*, 52(3), 387–412.

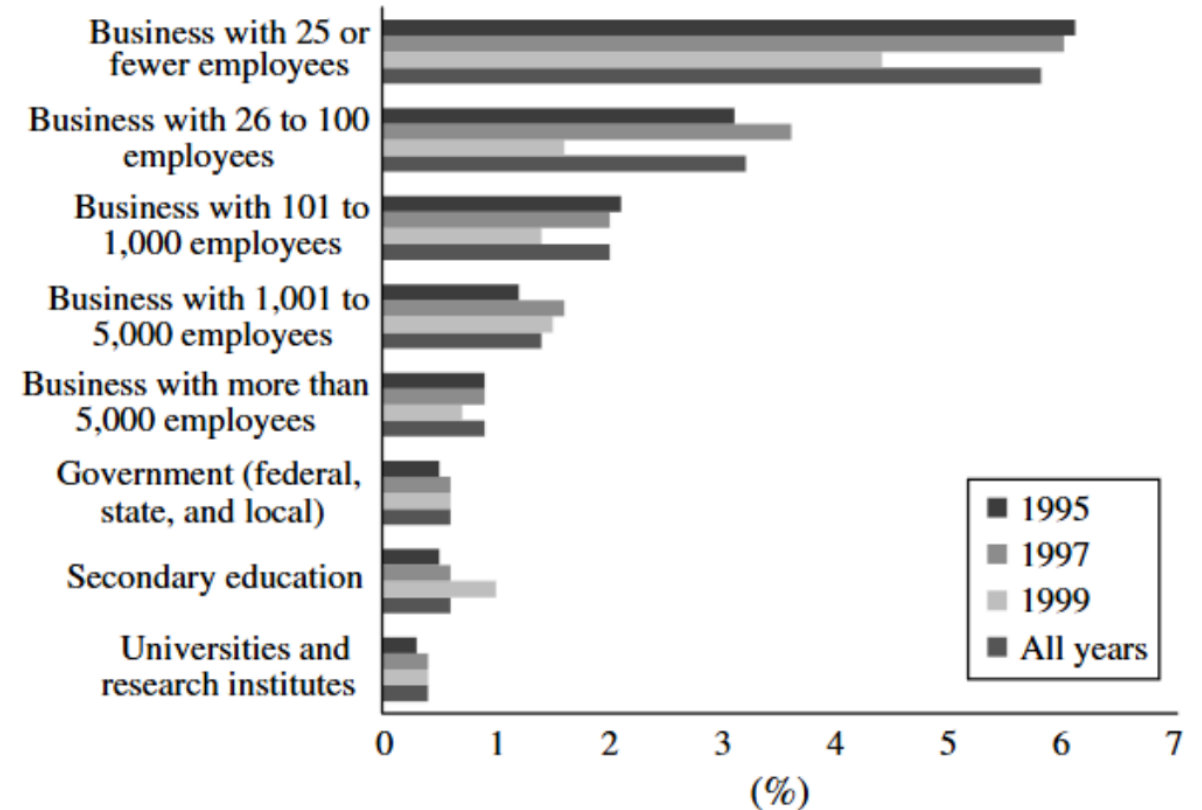
SMALL FIRMS GENERATE DISPROPORTIONATE NUMBERS OF FOUNDERS

Elfenbein et al. (2010) study U.S. scientists and engineers:

- Employees of small firms are much more likely to transition into self-employment
- For employees in firms with **1–25 workers**, entry is more than **three times** the sample average
- Prior small-firm employment is also associated with stronger early entrepreneurial outcomes

This is the *small-firm effect*.

Figure 1 Transitions into Self-Employment (Two Years Later) by Employer Type



Elfenbein, D. W., Hamilton, B. H., & Zenger, T. R. (2010). The small firm effect and the entrepreneurial spawning of scientists and engineers. *Management Science*, 56(4), 659–681.

ONE SMALL-FIRM PATTERN, FOUR POSSIBLE EXPLANATIONS

EXPLANATION	MECHANISM	MATRIX LOCATION
Preference sorting	People who value autonomy choose small firms and later entrepreneurship	SELECTION / NEUTRAL
Ability sorting	High-ability workers prefer stronger pay-for-performance and later exploit their ability	SELECTION / POTENTIALLY ENHANCING
Opportunity cost	Workers with less to lose leave paid work more readily	SELECTION / NEUTRAL
Entrepreneurial human capital	Broad tasks and proximity to founders teach workers how to build firms	TREATMENT / ENHANCING

Elfenbein et al. find evidence consistent with several mechanisms—not one pure treatment effect.

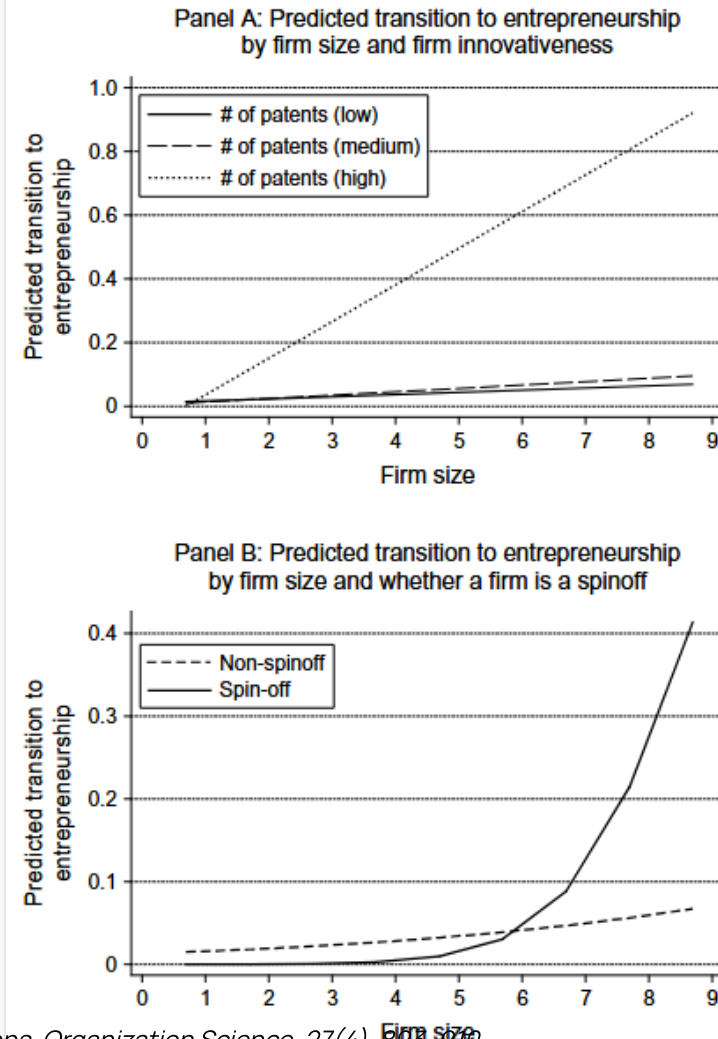
FIRM CLOSURE REMOVES THE INTERNAL CAREER OPTION

Kacperczyk & Marx revisit the small-firm effect in the automatic speech-recognition industry:

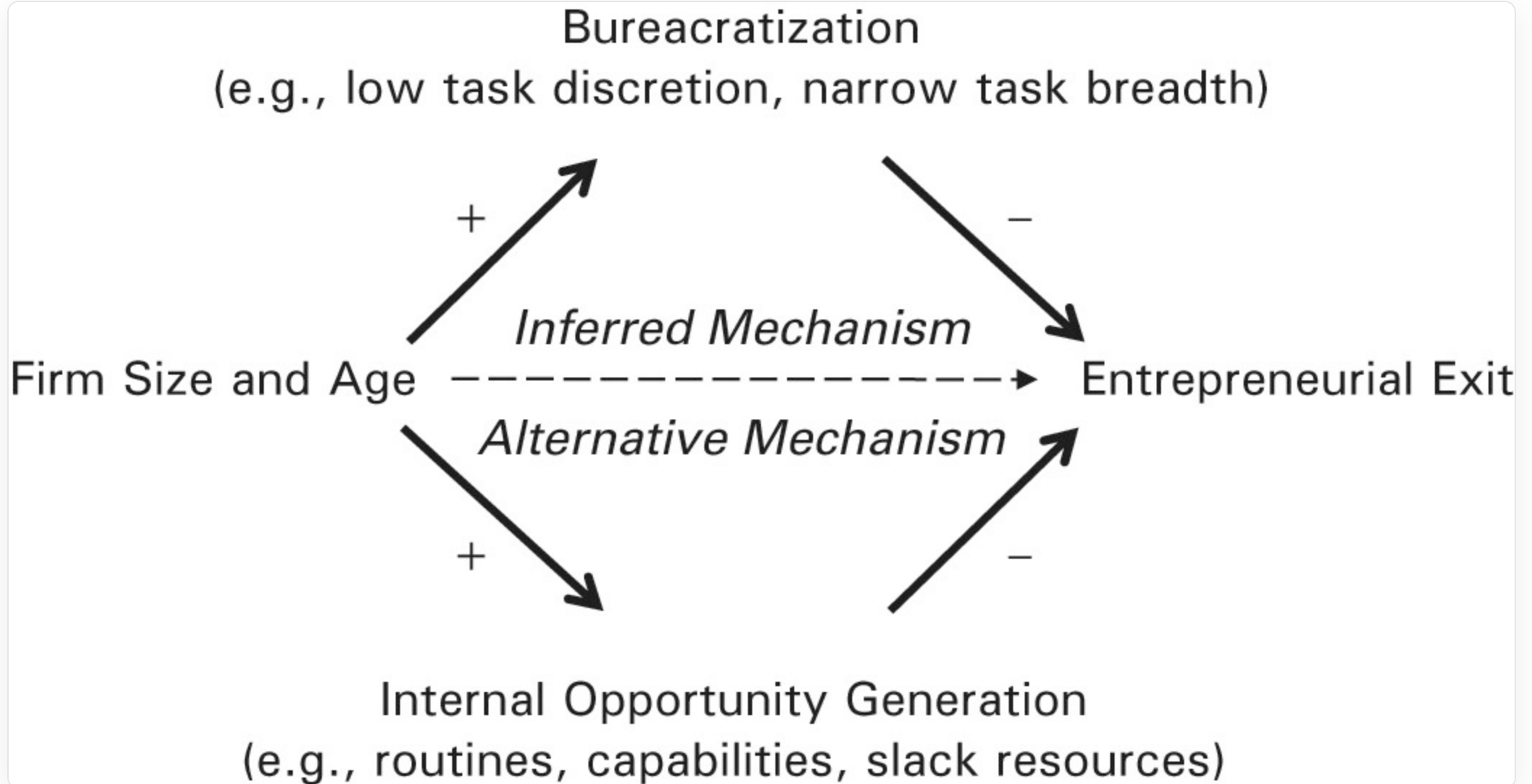
- Among **continuing** firms, smaller employers spawn more entrepreneurs
- At **firm dissolution**, internal opportunities disappear for everyone
- The relationship **reverses**: employees leaving larger dissolved firms are more likely to become entrepreneurs
- A one-standard-deviation increase in logged firm size is associated with about **15% higher odds** of entrepreneurship in the dissolution sample
 - If 5 in 100 leavers from an average-sized dissolved firm found a venture, then among leavers from a firm one SD larger it is roughly **5.7 in 100**

Kacperczyk, A., & Marx, M. (2016). Revisiting the small-firm effect on entrepreneurship: Evidence from firm dissolutions. *Organization Science*, 27(4), 893-910.

Figure 2 Interaction Effects for Predicted Transition to Entrepreneurship



ORGANIZATIONS CAN MOLD ENTREPRENEURS—AND KEEP THEM INSIDE



EMPLOYMENT CAN CREATE PORTABLE ENTREPRENEURIAL CAPITAL

WHAT EMPLOYEES LEARN	WHAT IT ALLOWS A FOUNDER TO DO
Industry and customer knowledge	Recognize valuable, feasible problems
Technical knowledge	Build or adapt products and processes
Regulatory and institutional knowledge	Navigate approval, reimbursement and legal constraints
Commercial knowledge	Price, sell, finance and scale
Managerial knowledge	Hire, coordinate and build routines
Relational knowledge	Identify whom to recruit, trust or approach

WHAT DO WORKERS LEARN?

Chatterji (2009) studies new medical-device ventures:

- **Spawns** founded by former employees of industry incumbents outperform other entrants (higher valuations, shorter time to funding)
- Their advantage does not appear to come mainly from copying the parent's patented technology
- Interviews instead emphasize regulatory strategy, reimbursement, marketing and managerial knowledge

However: talented employees may select into strong parent firms and later into entrepreneurship.

SELECTION

Founder ability → Parent employment → Spinoff performance

TREATMENT

Parent employment → Nontechnical knowledge → Spinoff performance

THE IDEA IS OFTEN DISCOVERED AT WORK

From a nationally representative survey in the US (so both growth and subsistence entrepreneurship):

- Most business ideas (**40%**) are “discovered” at work, next to personal use of a product
- Customer problems, technologies, market gaps and rejected internal projects can all become entrepreneurial opportunities
- Employment does more than pay a wage—it changes **what people are able to notice**

Table 5: Source of the business idea

	Freq.	%	Cum.
Identified a market opportunity through my work experience and this opportunity was specifically related to a supplier or customer of my employer	1,510	15.09	15.09
Identified a market opportunity through my personal use of an existing consumer product or recognizing an unmet need in a market I was familiar with	3,446	34.44	49.53
Read a media article which suggested a potential market opportunity	667	6.67	56.19
Partnered with another individual who had an existing idea	964	9.63	65.82
Other, please explain	833	8.32	74.15
Identified a market opportunity through my work experience and this opportunity was specifically related to my employer's business	2,587	25.85	100
Total	10,007	100	

PROFESSIONAL EXPERIENCE ALSO SHAPES *WHICH* IDEAS YOU DISCOVER

- Shane (2000) looked at startups that exploited 3D printing, invented at MIT. Despite the technology being well known at the time (patent applications, publications, industry consortium meetings), only **eight applications** were identified by the end of the 1990s
- None of these applications were obvious from knowledge about the technology *alone*

“All of the 3DP lab’s work was in one ceramic powder and one fluid, and medical applications require multiple fluids and multiple powders, specifically polymers. You couldn’t just look at the 3DP process and know that you could use it for medical purposes. To make use of the 3DP process for drug delivery, you had to know something about what drugs and drug delivery systems are made from and how drug manufacture operates.”

Walter Flamenbaum – Therics

Company	Entrepreneurs	Opportunity
Z Corp	Marina Hatsopoulos Walter Bornhorst Jim Bredt Tim Anderson	Manufactures a fast, inexpensive, office-compatible machine to make concept models for industrial and architectural design
Therics	Walter Flamenbaum	Manufactures pills with a superior microstructure through a fully integrated manufacturing process
Specific Surface	Mark Parrish Andrew Jeffrey	Manufactures ceramic filters for the power generation market in a one-step manufacturing process
Soligen	Yehorem Uziel	Manufactures machines to make ceramic molds for casting metal parts directly from a CAD model without wax forms or tooling
3D Partners	Andrew Kelly (and others)	Creates a service bureau to produce architectural models from CAD drawings
3D Orthopedics	Stephen Campbell	Provides a service to create artificial bone for weight bearing indications for use in surgery
3D Imaging	Lau Christianson Todd Jackson	Provides a service to create multicolor, three dimensional surgical models
Conferences	Michael Padnos	Establishes a chain of stores to make sculptures from photographs

RECOGNITION, NOT SEARCH

- The eight entrepreneurs described a discovery process that involved **recognition**, rather than a search for information. In none of the cases did an entrepreneur indicate that he or she was searching for the opportunity prior to its discovery
- Entrepreneurs are not better than others at searching for and discovering opportunities *in general*

*“If Bob Cohn, who has been responsible for drug delivery systems and product development at J&J had looked at it, it would have been different... [H]e would have turned around and said, ‘holy cow!’ This is a platform that’s wonderful for drug delivery systems. If you had showed it to someone who does tissue engineering, they would have said, ‘holy cow!’ This is wonderful for tissue healing matrices for our biological development of tissues and organs. **It’s a matter of mind set and background.**”*

Walter Flamenbaum – Therics

RECAP: ORGANIZATIONS AS FONTS OF ENTREPRENEURSHIP

Where you work and what you do influences your likelihood of becoming an entrepreneur—and how successful you will be. There are important trade-offs:

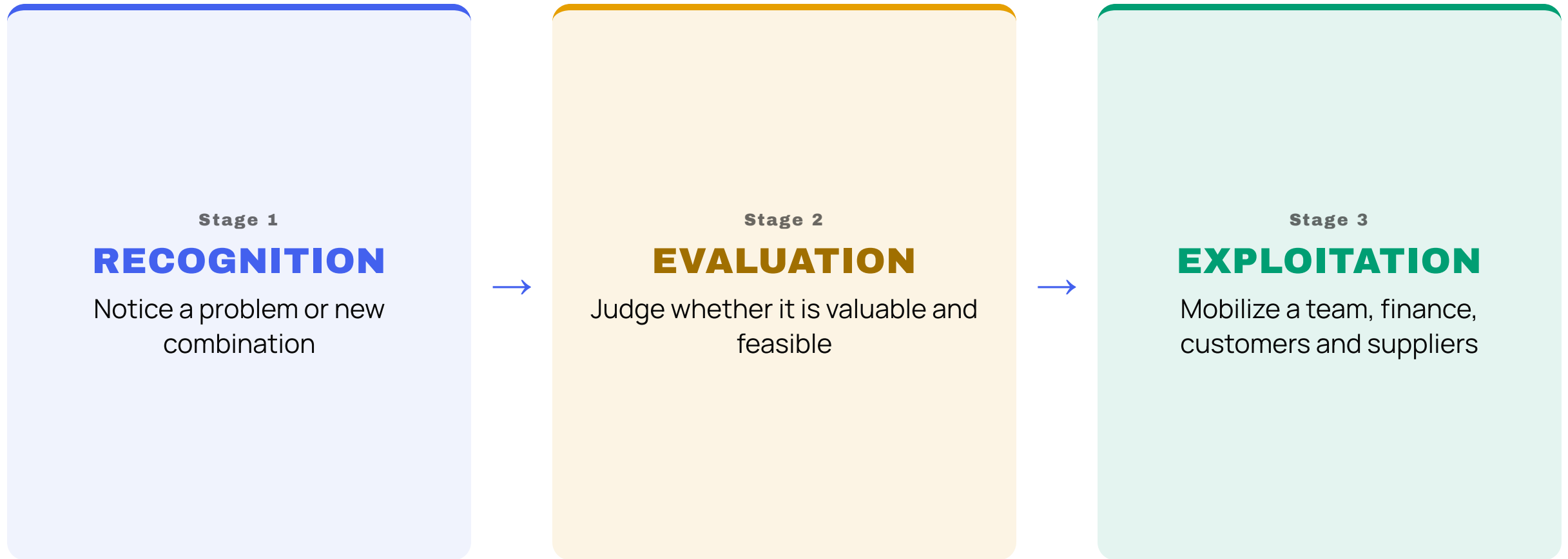
Large, innovative firm. Provides many resources and new technologies, but these companies are very good at retaining you—voluntarily through high opportunity costs, or involuntarily through non-competes, for example.

Large, bureaucratic firm. Offers fewer resources and less skill development useful in entrepreneurship, but these firms will care less if you want to spin off.

Small firm. Exposes you to many of the tasks entrepreneurs do, but you are less likely to be exposed to the technological frontier, and the chance of involuntary exit through firm failure is higher.

SOCIAL NETWORKS

AN OPPORTUNITY MUST BE RECOGNIZED, EVALUATED AND EXPLOITED



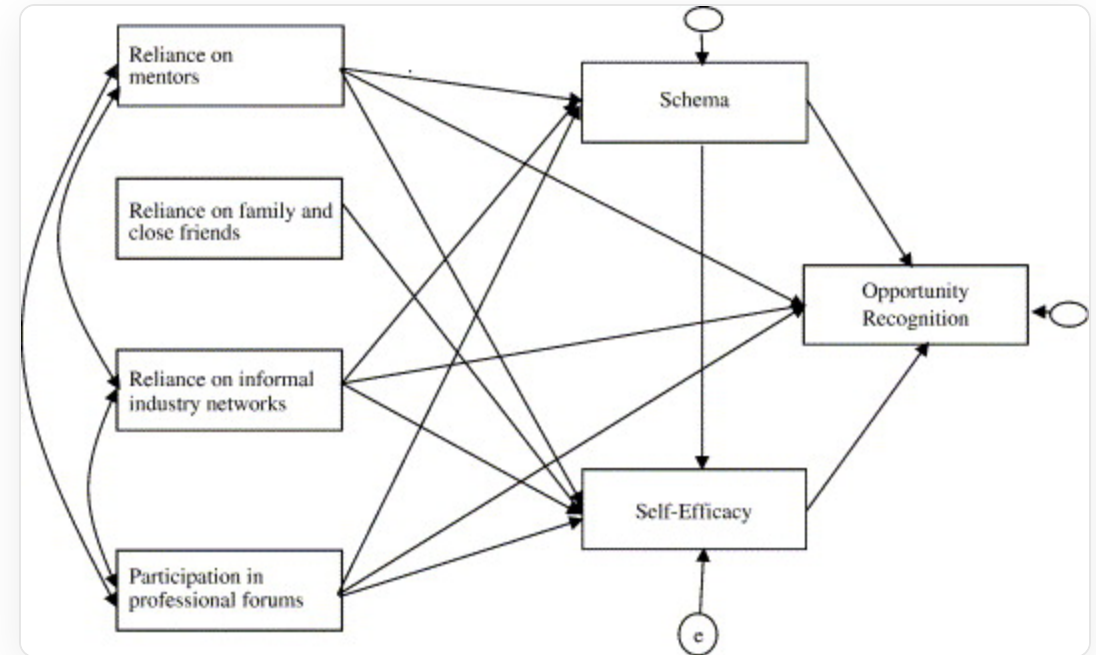
Networks can shape *all three stages*, but different relationships may be useful at each one.

OPPORTUNITY RECOGNITION HAS SOCIAL SOURCES

Ozgen & Baron (2007) study three sources of information:

- **Mentors** help interpret experience and build useful cognitive frameworks (“schemas”)
- **Professional forums** expose people to varied information and analogies
- **Informal industry contacts** reveal problems, practices and changing demand

These sources are positively associated with recognizing more opportunities.



THE IMPORTANCE OF SCHEMAS: OPPORTUNITY RECOGNITION AS PATTERN RECOGNITION

- Experienced entrepreneurs have a more clearly defined **prototype (schema)** of an opportunity
- Opportunity prototypes of experienced entrepreneurs are **richer in content** (they contain more dimensions)
- They focus more on factors related to **starting and running new ventures** (see table)
- Mentors might **substitute** for such entrepreneurial experience

Table 5 **Contrasting Prototypes of Novice and Experienced Entrepreneurs**

Discriminant profile: Novice entrepreneurs	Discriminant profile: Experienced entrepreneurs
How novel the idea is	Solving a customer's problems
Extent to which idea is based on new technology	Ability to generate positive cash flow
Superiority of product or service	Speed of revenue generation
Potential to change the industry	Manageable risk
Intuition or gut feel	Others in their network with whom to develop the venture

Baron, R. A., & Ensley, M. D. (2006). Opportunity recognition as the detection of meaningful patterns: Evidence from comparisons of novice and experienced entrepreneurs. Management Science, 52(9), 1331-1344.

ENTREPRENEURSHIP CAN SPREAD THROUGH PEERS

Two settings show social transmission of entrepreneurial entry:

WORKPLACE PEERS

NANDA & SØRENSEN (2010)

- Employees are more likely to become entrepreneurs when coworkers have prior entrepreneurial experience
- The association is strongest when people have **less entrepreneurial exposure elsewhere**
- Patterns are consistent with **information and practical know-how**, as well as **role models and demystifying failure**

UNIVERSITY PEERS

KACPERCZYK (2013)

- Among hedge-fund professionals, prior entrepreneurship among university peers predicts entry
- Effects are stronger for **nearby** and **same-gender** peers
- Consistent with **information exchange and social referencing** that reduces uncertainty

But neither study observes the mediators directly or isolates one mechanism.

Nanda, R., & Sørensen, J. B. (2010). Workplace peers and entrepreneurship. *Management Science*, 56(7), 1116–1126.

Kacperczyk, A. J. (2013). Social influence and entrepreneurship: The effect of university peers on entrepreneurial entry. *Organization Science*, 24(3), 664–683.

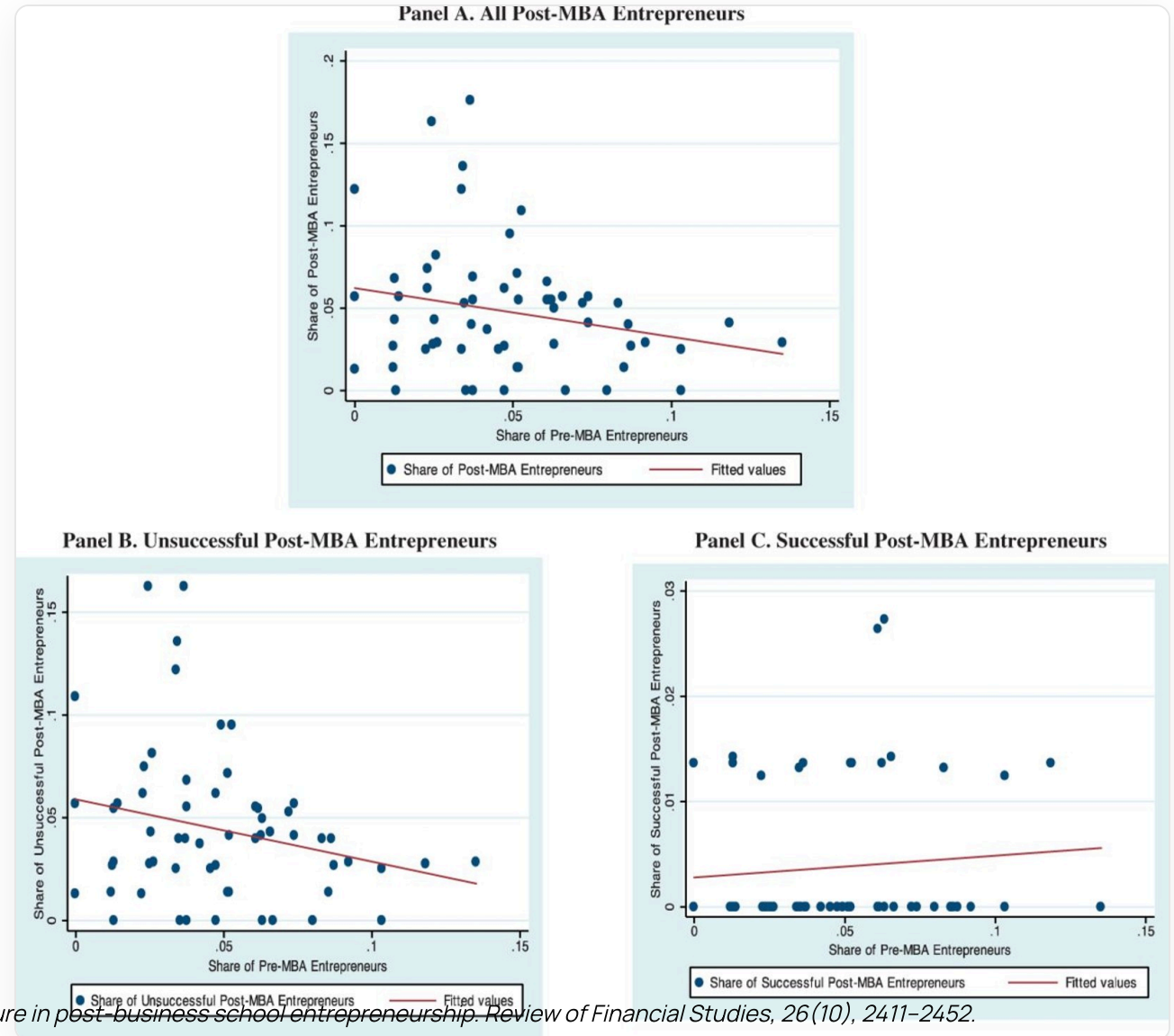
INFORMED PEERS MAY TALK YOU OUT OF A BAD STARTUP

Lerner & Malmendier exploit the assignment of Harvard MBA students to sections:

- Sections differed in the share of classmates with prior entrepreneurial experience
- More experienced entrepreneurial peers **reduced** later entrepreneurial entry
- The reduction came from **fewer unsuccessful ventures**
- The effect on successful ventures was substantially more positive

Better information can improve selection by preventing weak entry.

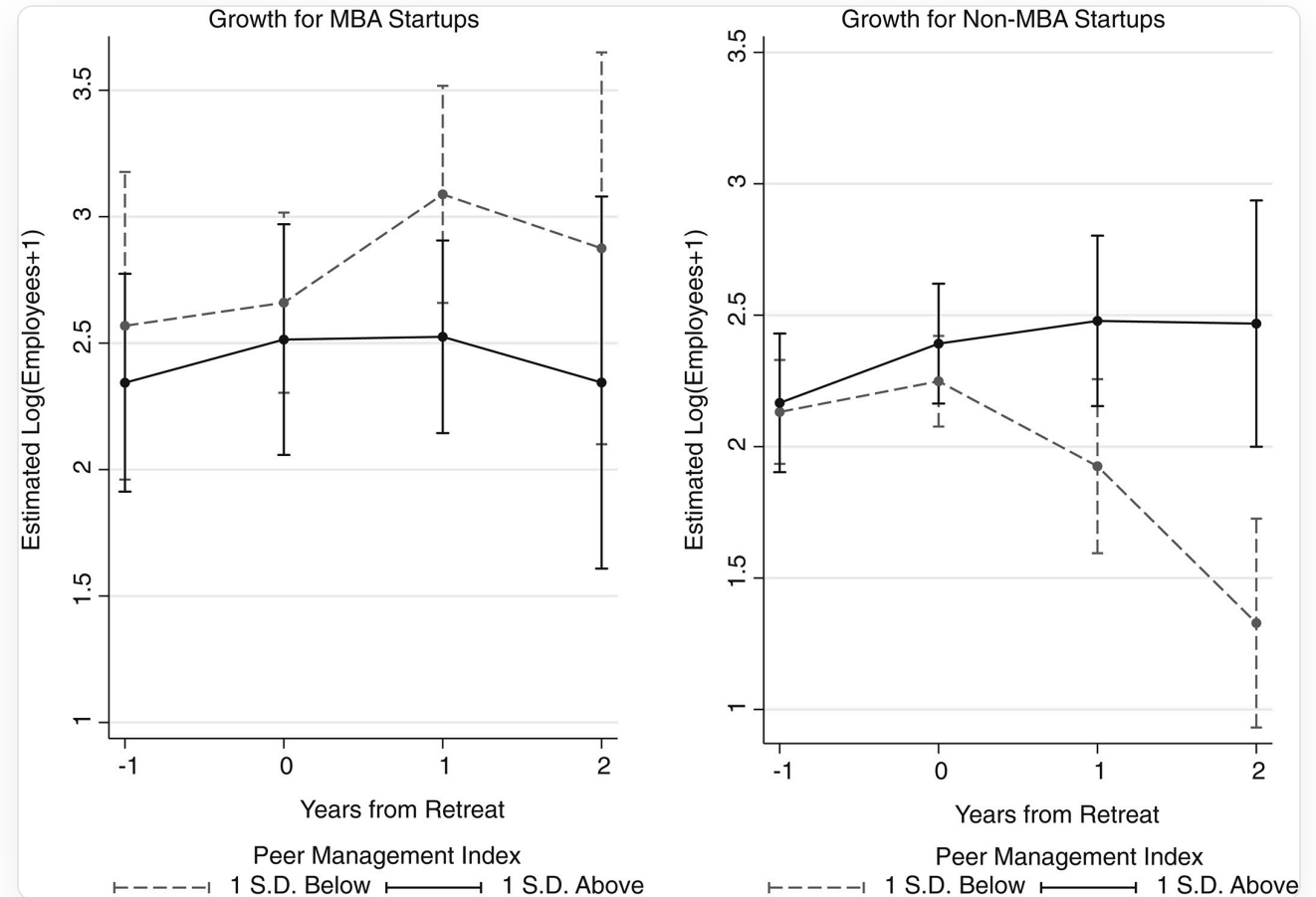
Lerner, J., & Malmendier, U. (2013). With a little help from my (random) friends: Success and failure in post-business school entrepreneurship. *Review of Financial Studies*, 26(10), 2411-2452.



ADVICE QUALITY MATTERS MORE THAN ADVICE QUANTITY

Chatterji et al. (2019) randomly matched 100 Indian technology startups with experienced peer advisers:

- Advisers differed in whether they used **active, formal people management**—regular meetings, goals and feedback
- Advice from active managers produced ventures that were **28% larger** after two years
- These ventures were **10 percentage points less likely to fail**
- Founders with an MBA or incubator/accelerator experience did *not* show the same response

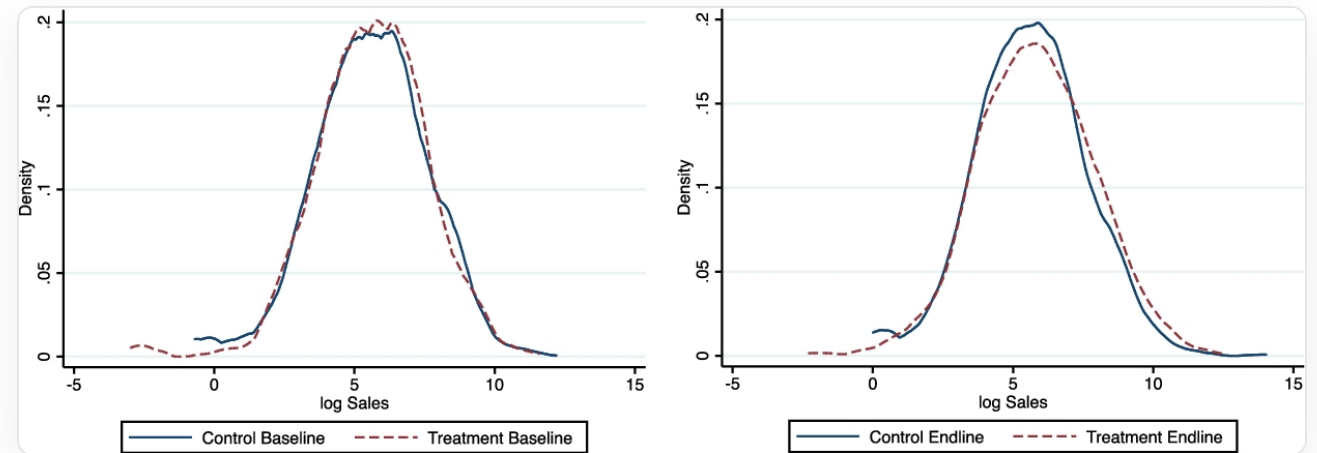


REPEATED PEER INTERACTION CAN IMPROVE FIRM PERFORMANCE

Cai & Szeidl randomized **2,820** young Chinese firms:

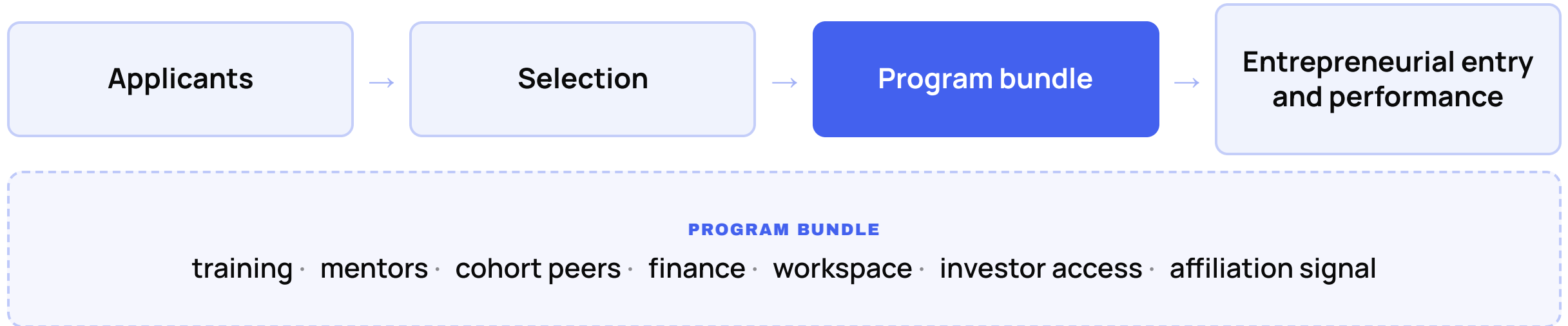
- Treated owner-managers met monthly in ten-person business groups for one year
- Meetings increased firm revenue by **8.1%** during the intervention
- Effects persisted one year after the meetings ended
- Profits, management practices, borrowing, clients and suppliers also improved

Additional interventions point to peer learning and supplier-customer matching as mechanisms.



INCUBATORS AND ENTREPRENEURSHIP PROGRAMS

AN INCUBATOR IS A BUNDLE—AND A SELECTION DEVICE



- Did the program **improve** the founder?
- Did it simply **admit** founders who would have succeeded anyway?
- Which **component** produced the outcome? For whom?

INCUBATORS ARE ASSOCIATED WITH MORE ENTREPRENEURSHIP

Lyons & Zhang evaluate a Canadian technology-entrepreneurship program combining education, mentors, cohort ties, investors and potential seed capital:

- They compare **335 accepted participants** with unaccepted applicants who reached the finalist stage
- Program participation is associated with **+24 percentage points** in later startup-related activity
- ...and **+23 percentage points** in later startup founding
- ...plus stronger venture survival, funding and employment outcomes



WORLD CLASS CURRICULUM

Learn how to build, launch and scale from successful entrepreneurs and award winning faculty.



ALUMNI COMMUNITY

Join a community of 1,000+ alumni and gain access to coaching, resources and big impact ventures.



INVESTMENT OPPORTUNITIES

Access NEXT Canada's investor network.



INCREDIBLE MENTORSHIP

Get the chance to be paired with a mentor who has successfully launched a venture and have them guide the way.



FREE OFFICE SPACE

You'll have access to a productive workspace surrounded by likeminded talented entrepreneurs.



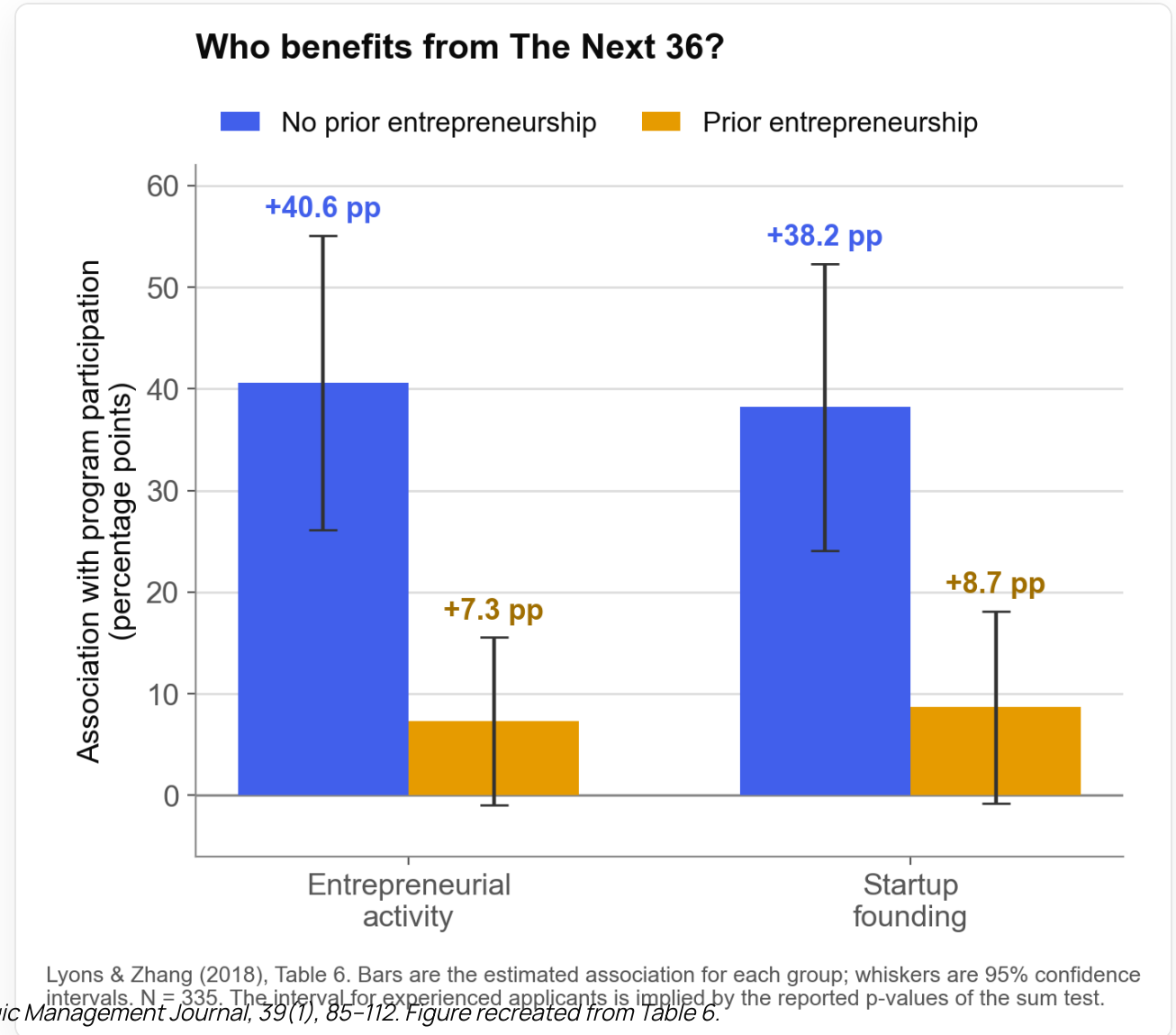
IN-KIND PRODUCTS AND SERVICES

Get the services your venture needs from our partners including cloud hosting, legal support and more.

THE LARGEST GAINS APPEAR AMONG TECHNOLOGY-FOUNDING NOVICES

For applicants without prior entrepreneurship, participation is associated with approximately:

- **+41 percentage points** in later entrepreneurial activity
- **+38 percentage points** in later startup founding
- Prior entrepreneurial experience reduces these estimated relationships by about **33** and **30** percentage points
- The reduction is concentrated among applicants who had already founded a **technology company**



THE PROGRAM APPEARS TO FILL KNOWLEDGE AND NETWORK GAPS

PARTICIPATION INCREASED

LYONS & ZHANG'S MECHANISM EVIDENCE

- Knowledge about evaluating and developing technology startups
- Access to investors and entrepreneurship-relevant relationships

LESS EVIDENCE OF CHANGE IN

WHAT DID NOT CLEARLY MOVE

- Preferences for entrepreneurship
- Risk tolerance
- Confidence

The program seems to expand what participants *know* and *whom they can reach*—not simply how entrepreneurial they feel.

CASH AND WORKSPACE ARE NOT THE SAME AS ENTREPRENEURIAL LEARNING

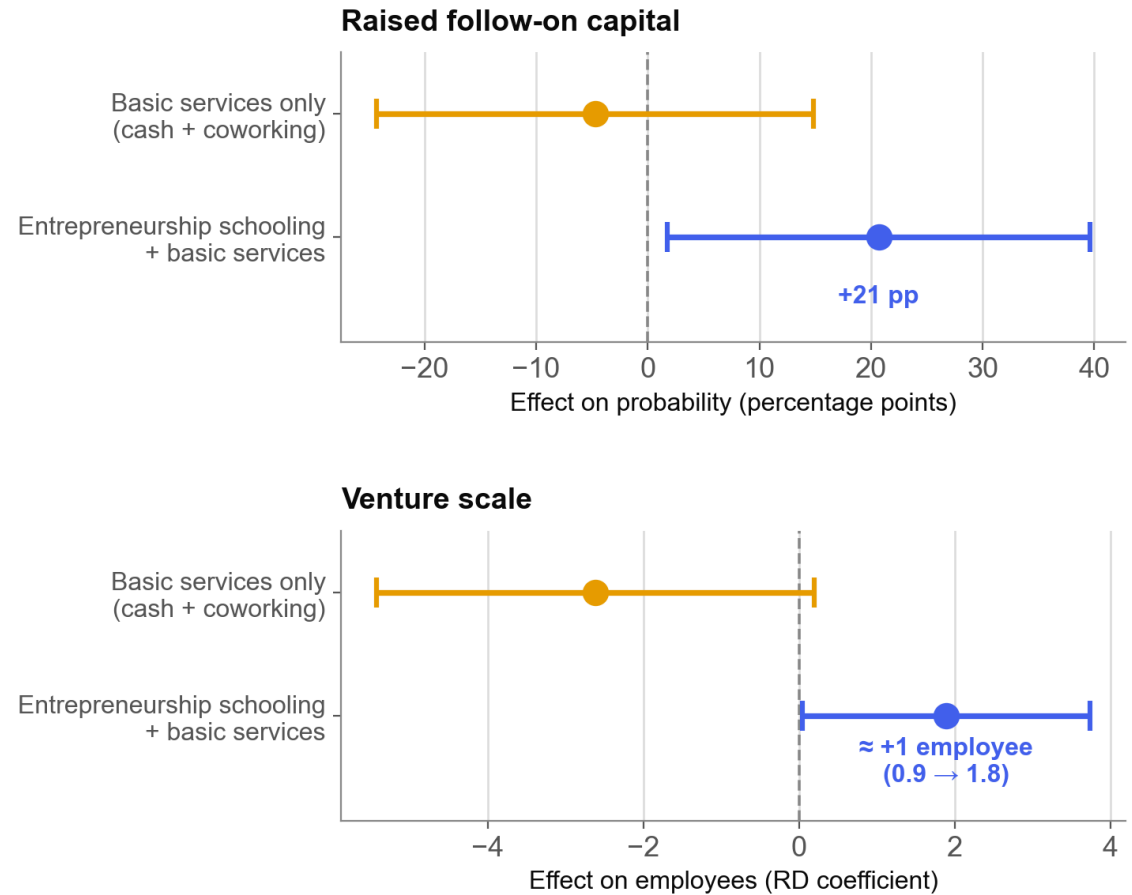
Gonzalez-Uribe & Leatherbee study Start-Up Chile using score cutoffs:

- **Basic services** combined cash and coworking space → no detectable effect on fundraising, employment or survival for ventures near the admission cutoff
- **Entrepreneurship schooling** bundled with those services → increased follow-on finance and venture scale

Structured learning—not the incubator label by itself—appears to be the active treatment.

Gonzalez-Uribe, J., & Leatherbee, M. (2018). The effects of business accelerators on venture performance: Evidence from Start-Up Chile. *Review of Financial Studies*, 31(4), 1566–1603. Figure recreated from Tables 6 and 7.

Start-Up Chile: regression-discontinuity estimates



Gonzalez-Uribe & Leatherbee (2018), Table 6 col. 4 and Table 7 col. 4 (p = 1, with controls). Whiskers are 95% confidence intervals.

SOME ACCELERATORS ACCELERATE LEARNING; OTHERS DO NOT

Hallen, Cohen & Bingham compare accepted and “almost accepted” ventures across early accelerators:

- Some programs substantially accelerate venture development
- Other programs show little evidence of doing so
- Sorting and signaling explain part—but not all—of the observed differences

Their qualitative evidence identifies a distinctive learning mechanism:

Broad, intensive and paced consultation

BROAD

Many perspectives from many advisers

INTENSIVE

Concentrated, repeated engagement

PACED

External deadlines and milestone sequences

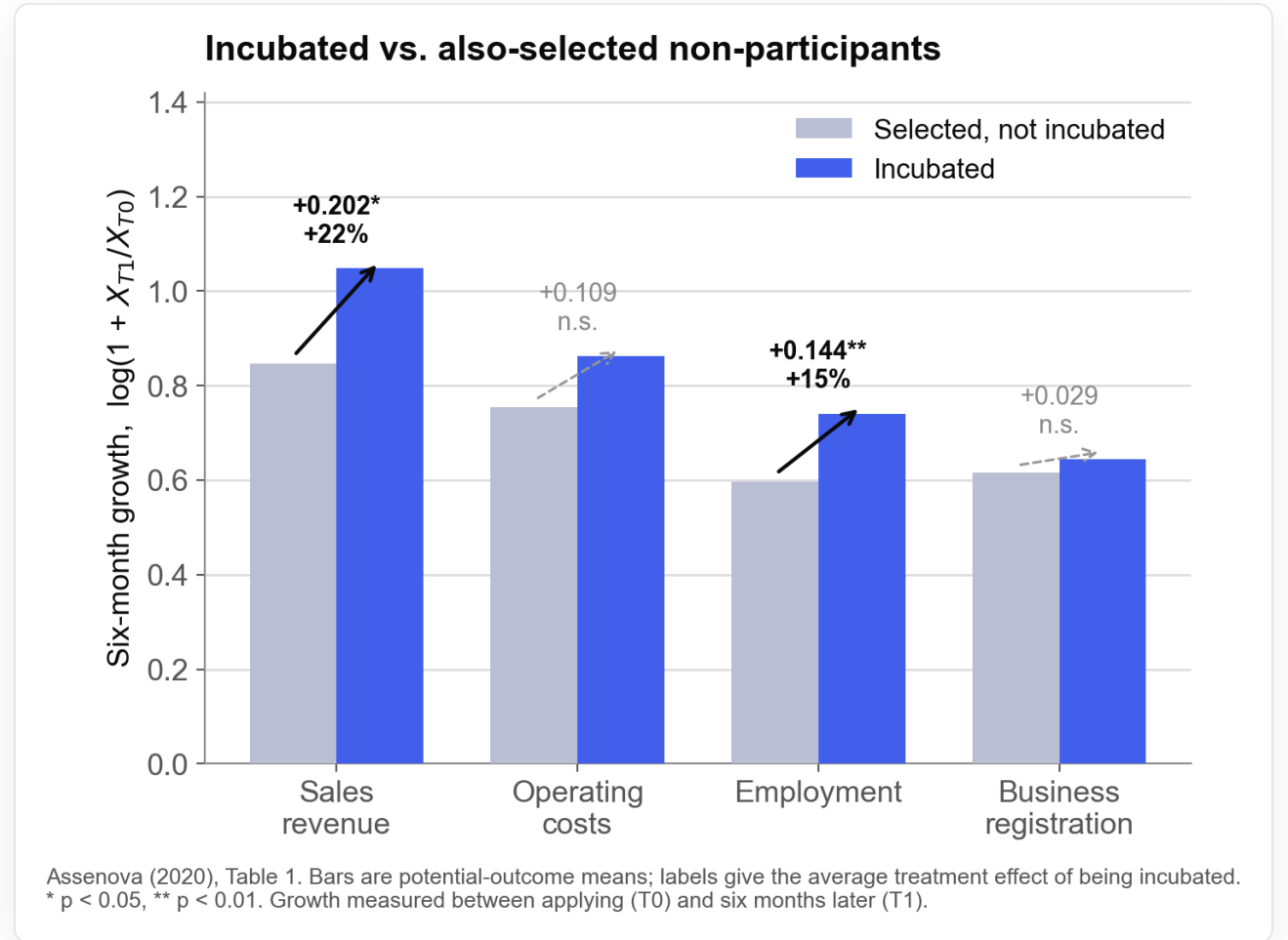
RESULT

Faster learning cycles

INCUBATION CAN HELP DISADVANTAGED ENTREPRENEURS SCALE

Assenova studies a South African incubator for socially and educationally disadvantaged entrepreneurs:

- Incubated firms are compared with similar, **also-selected but non-incubated** firms
- Over six months, incubated firms grew **22% more in revenue**
- They also grew **15% more in employment**
- The pattern suggests that incubation can help close gaps in prior knowledge and access



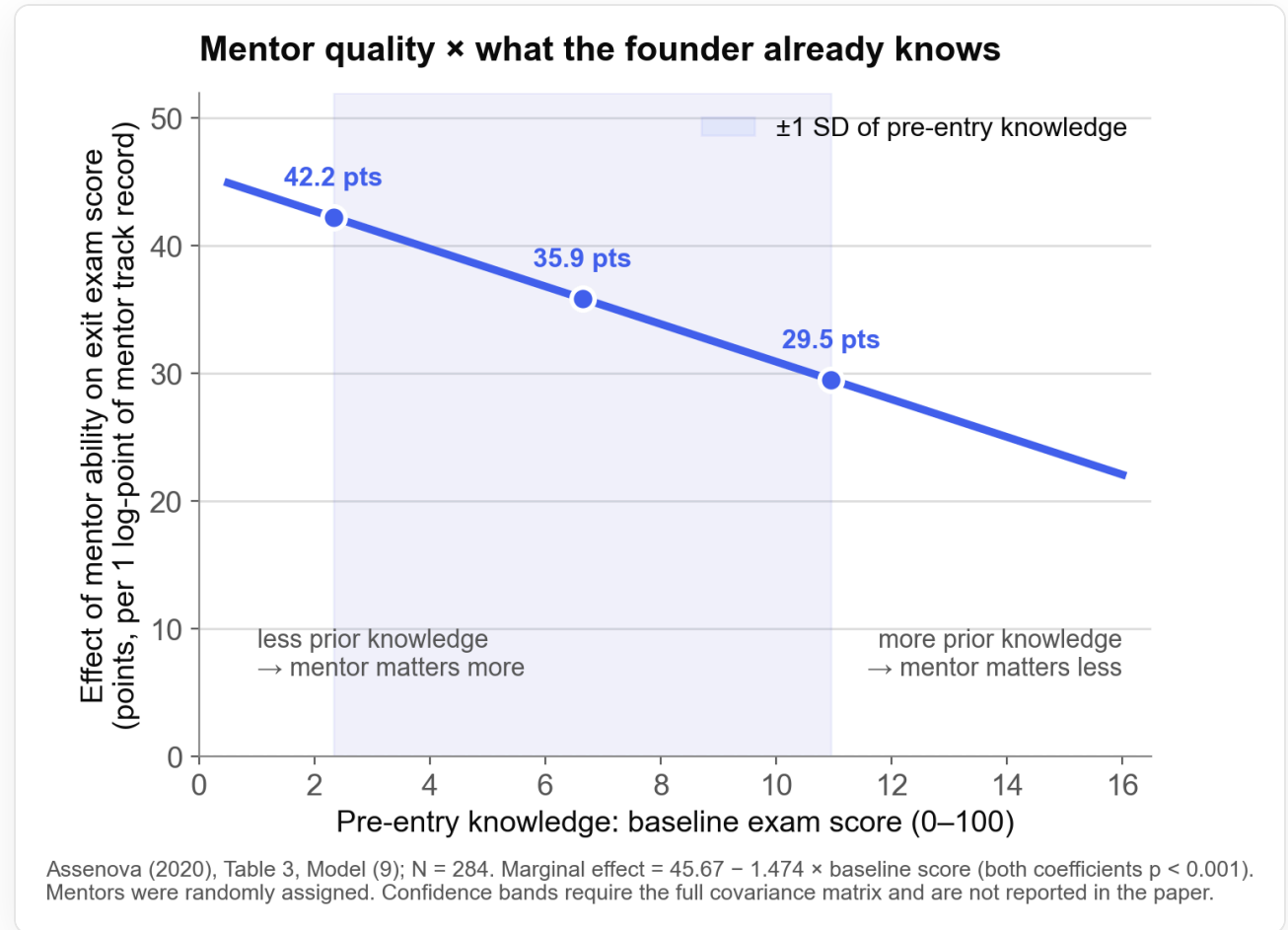
MENTOR QUALITY MATTERS MOST WHEN PRIOR KNOWLEDGE IS SCARCE

Within the same incubator, participants were **randomly assigned to mentors**:

- Assignment to a high-ability rather than low-ability mentor produced **3.2% higher revenue** one year later
- Profits were **3.5% higher**
- Mentoring generated more learning and larger performance improvements for founders with **less prior knowledge and experience**

The value of advice depends on the match between what the mentor provides and what the founder lacks.

Assenova, V. A. (2020). Early-stage venture incubation and mentoring promote learning, scaling, and profitability among disadvantaged entrepreneurs. *Organization Science*, 31(6), 1560–1578. Figure recreated from Table 3, Model (9).



CONTEXT IS MOST VALUABLE WHEN IT FILLS A GAP

EVIDENCE	WHO BENEFITS MOST?
Entrepreneurial coworkers (Nanda & Sørensen)	People with less exposure to entrepreneurship elsewhere
Active peer advice (Chatterji et al.)	Founders without overlapping MBA or accelerator experience
The Next 36 (Lyons & Zhang)	Applicants without prior technology-founding experience
High-quality mentors (Assenova)	Founders with less prior knowledge and experience

Support often *substitutes* for missing human and social capital rather than uniformly amplifying existing advantage.

ENTREPRENEURS DO NOT EMERGE FROM A VACUUM.

Organizations, networks and programs shape who enters
—and what they can do after entry.

Slides and the data underlying the graphs:

github.com/jmahieu/foundations-of-entrepreneurship